Ps 02

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1 Problem 11 DONE

1.1 Task

For sequences $f(n)$, $g(n)$, $h(n)$, and $k(n)$ prove that $f(n) = \Omega(h(n))$ and $g(n) = \omega(k(n))$ implies $f(n) + g(n) = \Omega(h(n) + k(n))$.

1.2 Solution

Assume $f(n) = \Omega(h(n))$, $g(n) = \omega(k(n))$. Show $f(n) + g(n) = \Omega(h(n) + k(n))$.

$$\Omega(h(n)) = \{f(n) : \exists c_1, n_1 > 0 : 0 \leq c_1 \cdot h(n) \leq f(n), \forall n \geq n_1\}$$

$$\omega(k(n)) = \{g(n) : \forall c_2 > 0, \exists n_2 > 0 : 0 \leq c_2 \cdot k(n) < g(n), \forall n \geq n_2\}$$

Let $n_0 = \max(n_1, n_2)$ and $c_2 = c_1$. Then $\forall n \geq n_0$:

$$\begin{gathered} 0 \leq c_1 \cdot h(n) \leq f(n) \\ 0 \leq c_1 \cdot k(n) < g(n) \\ 0 \leq c_1 \cdot (h(n) + k(n)) \leq f(n) + g(n) \end{gathered}$$

Hence $f(n) + g(n) = \Omega(h(n) + k(n))$. ■

2 Problem 12 DONE

2.1 Task

Let $f(n) = O(g(n))$ and $g(n) = o(h(n))$. Prove that $f(n) = o(h(n))$.

2.2 Solution

Assume $f(n) = O(g(n))$ and $g(n) = o(h(n))$. Show $f(n) = o(h(n))$.

$$O(g(n)) = \{f(n) : \exists c_1, n_1 > 0 : 0 \leq f(n) \leq c_1 \cdot g(n), \forall n \geq n_1\}$$

$$o(h(n)) = \{g(n) : \forall c_2 > 0, \exists n_2 > 0 : 0 \leq g(n) < c_2 \cdot h(n), \forall n \geq n_2\}$$

Let $c > 0$ be arbitrary. Choose $c_2 = \frac{c}{c_1}$ and $n_0 = \max(n_1, n_2)$. Then $\forall n \geq n_0$:

$$\begin{aligned} 0 \leq f(n) &\leq c_1 \cdot g(n) \\ &< c_1 \cdot c_2 \cdot h(n) \\ &< c_1 \cdot \frac{c}{c_1} \cdot h(n) \\ &< c \cdot h(n) \end{aligned}$$

Hence $f(n) = o(h(n))$. ■

3 Problem 13 DONE

3.1 Task

Use Stirling's formula

$$n! = \sqrt{2\pi n}\left(\frac{n}{e}\right)^n \left(1 + \Theta\left(\frac{1}{n}\right)\right).$$

to show that

(a) $n! = o(n^n)$,

(b) $n! = \omega(2^n)$,

(c) $\lg(n!) = \Theta(n \lg n)$.

3.2 Solution

3.2.1 (a)

$$\begin{aligned} \lim_{n\to\infty} \frac{\sqrt{2\pi n}\left(\frac{n}{e}\right)^n \left(1 + \Theta\left(\frac{1}{n}\right)\right)}{n^n} &= \lim_{n\to\infty} \sqrt{2\pi n} \cdot \frac{1}{e^n} \cdot \left(1 + \Theta\left(\frac{1}{n}\right)\right) \\ &= \lim_{n\to\infty} \sqrt{2\pi n} \cdot \frac{1}{e^n} \cdot 1 \\ &= 0 \end{aligned}$$

since $\sqrt{2\pi n}$ grows polynomially while e^n grows exponentially.

Hence $n! = o(n^n)$. ■

3.2.2 (b)

$$\begin{aligned}
\lim_{n\to\infty} \frac{\sqrt{2\pi n}\left(\frac{n}{e}\right)^n \left(1+\Theta\left(\frac{1}{n}\right)\right)}{2^n} &= \lim_{n\to\infty} \sqrt{2\pi n} \cdot \left(\frac{n}{2e}\right)^n \cdot \left(1+\Theta\left(\frac{1}{n}\right)\right) \\
&= \lim_{n\to\infty} \sqrt{2\pi n} \cdot \left(\frac{n}{2e}\right)^n \cdot 1 \\
&= \infty
\end{aligned}$$

Hence, $n! = \omega(2^n)$. ■

3.2.3 (c)

$$\begin{aligned}
\lg(n!) &= \lg\left(\sqrt{2\pi n}\right) + n\lg n - n\lg e + \lg\left(1+\Theta\left(\frac{1}{n}\right)\right) \\
\frac{\lg(n!)}{n\lg n} &= \frac{\frac{1}{2}\lg(2\pi n)}{n\lg n} + 1 - \frac{\lg e}{\lg n} + \frac{\lg\left(1+\Theta\left(\frac{1}{n}\right)\right)}{n\lg n} \\
\lim_{n\to\infty} \frac{\lg(n!)}{n\lg n} &= 0 + 1 - 0 + 0 = 1
\end{aligned}$$

Since the limit is a positive constant, $\lg(n!) = \Theta(n\lg n)$. ■

4 Problem 14 SKIP

4.1 Task

We define the iterated logarithm function

$$\lg^*(n) := \min\{i \in \mathbb{N} \mid \lg^{(i)}(n) \leq 1\}$$

where $\lg^{(1)}(x) := \lg x$, $\lg^{(k+1)}(x) := \lg(\lg^{(k)}(x))$. This function grows extremely slowly. Order the following functions by asymptotic growth:

#	Function
1	$\lg(\lg^*(n))$
2	$2^{\lg^*(n)}$
3	$\sqrt{2}^{\lg n}$
4	n^2
5	$n!$
6	$(\lceil \lg n \rceil)!$
7	$\left(\frac{3}{2}\right)^n$
8	n^3
9	$\lg^2 n$
10	$\lg(n!)$
11	2^{2^n}
12	$n^{1/\lg n}$
13	$\lg \lg n$
14	$\lg^* n$
15	$n \cdot 2^n$
16	$n^{\lg \lg n}$
17	$\lg n$
18	1
19	$2^{\lg n}$
20	$(\lg n)^{\lg n}$
21	e^n
22	$4^{\lg n}$
23	$(n+1)!$
24	$\sqrt{\lg n}$
25	$\lg^*(\lg n)$
26	$2^{\sqrt{2 \lg n}}$
27	n
28	2^n
29	$n \lg n$
30	$2^{2^{n+1}}$

4.2 Solution

5 Problem 15 DONE

5.1 Task

Use the substitution method to prove that the recursion

$$T(n) = 4T\left(\frac{n}{2}\right) + n$$

implies $T(n) = \Theta(n^2)$.

Hint: Subtract a lower order term.

5.2 Solution

- Upper bound: $T(n) = O(n^2)$

  Trying $T(n) \leq cn^2$ directly gives $cn^2 + n$, which doesn't close because the $+n$ term cannot be absorbed. To fix this, subtract a lower-order term from the hypothesis.

  - Inductive hypothesis: assume $T(k) \leq ck^2 - dk$ for all $k < n$.
  - Base case: $T(1) = a$. We need $a \leq c \cdot 1 - d \cdot 1 = c - d$. With $d = 1$, this requires $c \geq a + 1$, which is satisfiable.
  - Substitute:

$$\begin{aligned} T(n) &\leq 4\left(c \cdot \frac{n^2}{4} - d \cdot \frac{n}{2}\right) + n \\ &= cn^2 - 2dn + n \\ &= cn^2 - dn - (dn - n) \\ &\leq cn^2 - dn \end{aligned}$$

  Last step holds when $dn - n \geq 0$, i.e. $d \geq 1$. Pick $d = 1$.
  Since $cn^2 - dn \leq cn^2$, this establishes $T(n) = O(n^2)$.

- Lower bound: $T(n) = \Omega(n^2)$

  - Inductive hypothesis: assume $T(k) \geq ck^2$ for all $k < n$.
  - Base case: $T(1) = a$. We need $a \geq c \cdot 1 = c$, so choose $c \leq a$.
  - Substitute:

$$\begin{aligned} T(n) &\geq 4\left(c \cdot \frac{n^2}{4}\right) + n \\ &= cn^2 + n \\ &\geq cn^2 \end{aligned}$$

  Holds trivially since $n > 0$.

- Conclusion

  Since $T(n) = O(n^2)$ and $T(n) = \Omega(n^2)$, we have $T(n) = \Theta(n^2)$. ■

6 Problem 16 DONE

6.1 Task

For the recurrence

$$T(n) = 4T\left(\frac{n}{3}\right) + n$$

sketch the recursion tree and derive a guess for an asymptotic upper bound of its solution. Then use induction to prove your guess.

6.2 Solution

- Deriving the guess via recursion tree

  The recurrence $T(n) = 4T(n/3) + n$ spawns 4 subproblems of size $n/3$ at each level. At level k there are 4^k nodes, each contributing $n/3^k$ work, so the total work at level k is $(4/3)^k \cdot n$. The tree has depth $\log_3 n$. Since the ratio $4/3 > 1$, the geometric series is dominated by its last term:

  $$4^{\log_3 n} = n^{\log_3 4}$$

  This suggests $T(n) = O(n^{\log_3 4})$.

- Proof by substitution

  - Inductive hypothesis: assume $T(k) \leq ck^{\log_3 4} - dk$ for all $k < n$.
  - Base case: $T(1) = a$. We need $a \leq c \cdot 1 - d \cdot 1 = c - d$. With $d = 3$, this requires $c \geq a + 3$, which is satisfiable.
  - Substitute:

  $$\begin{aligned} T(n) &\leq 4\left(\frac{n^{\log_3 4}}{4} - d \cdot \frac{n}{3}\right) + n \\ &= cn^{\log_3 4} - \frac{4dn}{3} + n \\ &= cn^{\log_3 4} - dn - \left(\frac{dn}{3} - n\right) \\ &\leq cn^{\log_3 4} - dn \end{aligned}$$

  Last step holds when $\frac{dn}{3} - n \geq 0$, i.e. $d \geq 3$. Pick $d = 3$.

Since $cn^{\log_3 4} - dn \leq cn^{\log_3 4}$, this establishes $T(n) = O(n^{\log_3 4})$. ■

7 Problem 17 DONE

7.1 Task

Consider the recursion describing Strassen's algorithm for matrix multiplication. Using the Master Theorem, prove that its running time $T(n)$ satisfies $T(n) = O(n^{\lg 7})$.

7.2 Solution

Strassen's algorithm satisfies the recurrence $T(n) = 7\,T(n/2) + \Theta(n^2)$.

Apply the Master Theorem with $a = 7$, $b = 2$, $f(n) = \Theta(n^2)$.

We have $n^{\log_b a} = n^{\log_2 7} = n^{\lg 7}$. Since $\lg 7 \approx 2.81$, the cost $f(n) = n^2$ is polynomially smaller than $n^{\lg 7}$:

$$f(n) = O(n^{\lg 7 - \epsilon}) \quad \text{with } \epsilon \approx 0.81$$

This is Case 1 of the Master Theorem, which gives $T(n) = \Theta(n^{\lg 7})$, and therefore $T(n) = O(n^{\lg 7})$.
■